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Nexus Fraud Intelligence

Nexus

Meet the Team



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Summary

Problem & Solution



The Problem

Traditional fraud detection analyzes transactions individually, making it difficult to identify organized fraud networks.



Our Solution

Nexus uses **Graph Intelligence** and **Generative AI** to uncover hidden fraud networks and generate intelligent investigation reports.



The Impact

Reduce financial losses, accelerate investigations, and improve fraud response.



DATA SOURCES

Powering Nexus Fraud Intelligence



Data Sources

- IEEE-CIS Fraud Detection
- PaySim Financial Transactions

These public datasets are widely used in fraud detection research and provide realistic transaction scenarios.



Current Prototype

- Mock transaction data
- Simulated fraud network graph

For the 30% prototype, we used mock data to validate the workflow and core functionalities of the platform.



Why These Datasets?

- Realistic fraud scenarios
- Large-scale transactions
- Suitable for fraud pattern analysis

These datasets provide the ideal foundation for training robust AI models and detecting complex fraud patterns.



In Short:

For the 30% prototype, we used mock data to validate the workflow. **IEEE-CIS and PaySim are selected for the next training phase.**

TECH STACK

Technologies powering Nexus Fraud Intelligence



Generative AI

LLM-based report generation (planned integration)



Graph Intelligence

Cytoscape.js +
Graph Analysis



Frontend

React + Vite



Prototype Data

Mock JSON Data used
for prototype and
validation



Visualization

Interactive fraud
network graph for
pattern discovery



Future Backend

FastAPI +
PostgreSQL



OUR GOAL

Build a scalable AI-powered fraud intelligence platform using graph analysis and generative AI to help financial institutions detect, investigate, and prevent fraud effectively.

Project Overview

AI-Powered Financial Fraud Intelligence Platform

What Nexus Does



Detects hidden fraud networks



Connects suspicious relationships



Generates AI investigation reports



Speeds up fraud investigations



Reduces financial losses



Data Pipeline & Utilization

How Data Flows



Input



Accounts



Transactions



Device IDs



Process



Graph Intelligence



Generative AI



Output



Fraud Risk Score



Investigation Report

Why This Idea?



Solves a real **financial fraud challenge**



Leverages **Generative AI** for investigation reports



Uses **Graph Intelligence** to uncover hidden fraud networks



Supports **faster and smarter** fraud investigations



Designed for **real-world banking** applications



Summary



AI-powered fraud intelligence platform



Detects hidden fraud networks



Uses Graph Intelligence & Generative AI



Generates AI investigation reports



Supports faster and smarter fraud response



Prototype Validation

We have built the foundation of Nexus and validated the core components of our prototype.



Public datasets selected (**IEEE-CIS & PaySim**)



Data pipeline designed and validated



Graph relationships successfully modeled



AI investigation report prototype completed



Ready for model training and **dashboard integration**



Nexus Fraud Intelligence

AI-powered fraud network detection prototype

Prototype 30%

Transactions

24,512

High Risk Alerts

128

Fraud Networks

14

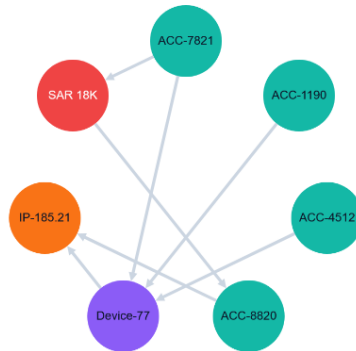
AI Reports

37

Fraud Network Graph

Accounts, devices, IPs, and transactions connected together.

Generate AI Report



Risk Analysis

Fraud Risk Score

96%

Critical Risk

Shared device across 4 accounts

Multiple transfers within 3 minutes

Same IP used by suspicious accounts

Recent Alerts

Case ID	Account	Amount	Signal	Risk	Status
#1021	ACC-7821	SAR 18,000	Shared Device	High	Under Review
#1022	ACC-1190	SAR 9,400	Same IP	Medium	Open
#1023	ACC-4512	SAR 2,100	Unusual Time	Low	Closed



Challenges & Future Roadmap



Challenges

- Limited access to real banking datasets.
- Privacy and regulatory compliance.
- Integrating data from multiple financial systems.



Support Needed

- Access to anonymized banking datasets.
- Mentorship from fraud and AI experts.
- **Banking APIs** or financial technology partnerships.
- Industry feedback for model validation.



Future Roadmap

- Train and optimize the AI model.
- Develop an interactive investigation dashboard.
- Integrate all platform components.
- Validate the solution with real-world scenarios.



Our Goal

Build an **AI-powered** fraud intelligence platform that combines **Graph Intelligence** and **Generative AI** to help financial institutions **detect organized fraud faster**, reduce financial losses, and **improve investigation efficiency**.



Building the Future of Financial Fraud Intelligence

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Thank you